

PhD Research Plan

Doctoral programme: Languages, Literatures and Cultures, and Their Applications **Department:** Department of Applied Linguistics — Universitat Politècnica de València (UPV) **PhD applicant:** Ana Eslava-Graterol **Director:** Prof. M^a Luisa Carrió-Pastor (Dept. of Applied Linguistics, UPV) **Co-director:** Prof. Ricardo Casañ-Pitarch (Dept. of Applied Linguistics, UPV) **Date:** 08/08/2026 · **Mode / duration:** [full-time 3 yr / part-time 5 yr — to confirm]

Reader's guide. This plan follows the UPV nine-section template. It develops one environment — **MLLDE (Mobile Language Learning Design Environment)**: a design-and-measurement environment for mobile-native language-learning applications — apps built phone-first (offline-capable PWAs, short-form lessons, conversational AI tutoring), not desktop e-learning shrunk to a screen. MLLDE advances two contributions: ① **writing / AI literacy** (extending Objective 4 of the applicant's MA thesis, instantiated in Academic Writing Journey) and ② **speaking / pronunciation** (Architecture Voice Design + a CEFR-aligned, acoustically-measurable Measurement Module in xAPI). A mobile-native gamified prototype is MLLDE's first working demo. Live artefacts already produced are linked in §3.3.

1. Title

Teaching Language in the Age of LLMs — The Mobile Language Learning Design Environment (MLLDE): an ethical, AI-literate design-and-measurement framework for mobile-native language apps, developing writing and speaking.

2. Abstract

Large language models (LLMs) have changed how people write, speak, and learn languages, and the smartphone is where most informal language study now happens. Yet mobile language apps optimise for streaks over the ability to act in the language; LLM-generated academic text is structurally fluent but interpersonally flat (weak stance, engagement, and self-mention), so learners who rely on it uncritically risk a homogenised, voiceless register; and AI speaking assessment, though technically mature, produces evidence that cannot be exchanged or accumulated across systems. Teachers hold robust pedagogy but lack a principled, ethical, and measurable way to turn it into engaging, phone-first learning that also builds learners' AI literacy (a study-and-employment skill, SDG 4.3–4.4).

This thesis proposes and empirically validates the **Mobile Language Learning Design Environment (MLLDE)**: a replicable, CEFR-anchored design-and-measurement environment for mobile-native language applications that integrates responsible LLM conversational tutoring with curriculum design and treats AI literacy as a designed outcome. MLLDE advances two contributions: ① **writing / AI literacy**, instantiated in Academic Writing Journey (AWJ) — the direct continuation of Objective 4 of the applicant's MA corpus study — teaching learners to retain rhetorical agency against the LLM signature; and ② **speaking / pronunciation**, comprising Architecture Voice Design (a CEFR-anchored architecture that extends the AWJ language-based custom GPT *PhD Chema* — whose first version does not assess pronunciation — by adding pronunciation as a first-class, assessed channel via ASR/GOP) and a Measurement Module that formalises acoustic pronunciation evidence as an interoperable, CEFR-aligned xAPI profile (IEEE 9274.1.1-2023) with dual telemetry logging both the learner's performance and the AI's teaching act. The Measurement Module follows a **Measure ≠ Decide** principle — the model observes, a deterministic engine assigns the CEFR band, so nothing on screen is invented — and an **evidence-honesty policy in three tiers**: Tier 1, a CEFR rubric over the ASR transcript (already in production); Tier 2, instrumented (vendor) pronunciation scoring; and Tier 3, a research-grade custom Goodness-of-Pronunciation model (the thesis's deep contribution) — so that no report ever claims more precision than the system holds (an unmeasured skill shows "—", never an invented value). A mobile-native gamified prototype (offline PWA, WCAG-AA, with an answer-withholding AI mentor) is the environment's first mobile-native demo. General objective: to develop, formalise, and validate MLLDE and its design principles. Methodology: Design-Based Research operationalised through the Successive Approximation Model, with mixed-methods evaluation of learning, AI-literacy and pronunciation gains, engagement, and replicability across CEFR levels and target languages.

3. Background and state of the art

3.1 The problem (three converging needs). - Mobile-native design. Smartphones are the primary gateway to informal study, but dominant apps privilege decontextualised recognition and retention metrics over communicative/pragmatic competence; mobile learning needs its own design logic rather than shrunken e-learning (Clothier, 2026; Kukulska-Hulme & Shield, 2008; Stockwell, 2010; Godwin-Jones, 2011). - **Writing / AI literacy.** LLM academic text over-produces textual scaffolding (logical and frame markers) while under-producing interactional resources (hedges, boosters, attitude, engagement, self-mention) and fabricates evidentials (Jiang & Hyland, 2024, 2025a–c; Reinhart et al., 2025; Kobak et al., 2025; Walters & Wilder, 2023). Detectors are brittle and biased against non-native writers (Liang et al., 2023; Erol et al., 2025); the responsible reply is pedagogical, not forensic — universities that once built academic literacy must now build AI literacy (Almatrafi et al., 2024; Koltay, 2025; Kong et al., 2023; Cotton et al., 2023). - **Speaking / pronunciation.** AI speech technologies (ASR, forced alignment, Goodness-of-Pronunciation; Witt & Young, 2000) can score phoneme accuracy, fluency, prosody, and intelligibility, and the CEFR Companion Volume (Council of Europe, 2020) frames pronunciation as phonological competence; but acoustic metrics, CEFR descriptors, and learning-analytics standards (xAPI) remain disconnected, so pronunciation evidence is not interoperable across intelligent tutors, LMSs, and Learning Record Stores. This need is industry-driven, not only theoretical: the requirement for a measurable, xAPI-based voice/pronunciation mode arose from an authentic client request in the applicant's professional practice, grounding this contribution in real demand and in the Design-Based Research tradition of authentic contexts.

3.2 State of the art (by strand). AI & academic literacy (Almatrafi et al., 2024; Koltay, 2025; Kong et al., 2023). Metadiscourse & the human/LLM contrast (Hyland, 2005; Hyland & Tse, 2004; Ädel, 2006; Jiang & Hyland, 2024–2025c). MALL & microlearning design (Kukulska-Hulme & Shield, 2008; Stockwell, 2010; Godwin-Jones, 2011; Clothier, 2026). Gamification, CLIL, task-based pedagogy (Deterding et al., 2011; Casañ-Pitarch's Gamified-CLIL/IPO; Coyle et al., 2010; Harmer, 2007; Johnson, 2002; Crawford, 2001). Cognition, lexis, pragmatics (Mayer, 2009; Lewis, 1993; Kasper & Rose, 2001). Automatic pronunciation assessment (Witt & Young, 2000; Council of Europe, 2020). Learning analytics & interoperability (xAPI / IEEE 9274.1.1-2023; ADL, 2023). Responsible AI & regulation (EU AI Act Art. 50; European Accessibility Act; GDPR; UDL/CAST).

Synthesis gap. These strands are rarely integrated into one mobile-native method that is built for AI-literate language learning, authorable by educators, and measurable through interoperable analytics for both writing and speaking. MLLDE addresses this gap.

3.3 Previous results of the team and the applicant. The Department of Applied Linguistics at UPV provides the direct scientific foundation: Prof. Carrió-Pastor's work on metadiscourse and the METOOL tagging tool (Carrió-Pastor, 2020), and Prof. Casañ-Pitarch's work on Gamified-CLIL, the Input–Processing–Output (IPO) model, and LLMs in language teaching. On this basis the applicant has already produced, and made openly available, a working lineage of integration — Academic Writing Journey (first model) → a mobile-native prototype → MLLDE (the generalised method):

Artefact	Stage in MLLDE	Link
MA thesis — metadiscourse in TFM abstracts	Empirical baseline (contribution ①)	[RiuNet — to add]
Academic Writing Journey — instructional design (Canva)	Design	https://profeanaeslava.my.canva.site/academicwritingjourney
Academic Writing Journey — gamified HTML course	First implementation	https://profeanaeslavaedtech.github.io/Gamification/
The Abstract Heist — escape-room module	Gamified assessment	https://profeanaeslavaedtech.github.io/escaperoom/
Abstract Generator / hosted prototype (UPV)	Output-stage tool	https://personales.alumno.upv.es/aeslgra/
Interactive Corpus Data Book (metadiscourse)	Data visualisation	https://profeanaeslavaedtech.github.io/portofolio/meta-discourse-visualization.html

Artefact	Stage in MLLDE	Link
Mobile-native gamified prototype (MLLDE demo)	First mobile-native demo	[private prototype]

AWJ was independently rated 10/10 for inclusive eLearning design by UPV (2026) with zero learner navigation errors, and proved platform-agnostic (rebuilt Canva → Articulate 360 Rise, SCORM/xAPI; delivered across four LMS families). AWJ already integrates the custom-GPT tutor *PhD Chema* — a language-based conversational agent — across all six levels; its first version does not assess pronunciation acoustically, which is exactly the gap the PhD's Architecture Voice Design and acoustic Measurement Module fill. A mobile-native gamified prototype — offline PWA, WCAG-AA, with an answer-withholding AI mentor (scaffolds under explicit consent; never does the learner's task) — has been built as a working demo evidencing that the Output stage of the IPO model (Casañ-Pitarch) is feasible on the phone.

The Measurement Module is, moreover, already **live in an applied production instance: TK Coach ("Capi", Talketika)**, a business-English voice tutor, runs **Tier 1** end-to-end — three observing LLM agents (tutor · corrector · evaluator) plus a **deterministic code consolidator** (not a fourth LLM) that computes the 0–10 CEFR band and emits a single xAPI statement to a conformant LRS (Veracity), under an **acoustic gate** (turns below ASR-confidence 0.70 are logged as incidents, never scored) and a strict separation of a **measurement stream** (once per session) from a **pedagogy stream** (per turn, which never scores). This provides real-use evidence of feasibility for the speaking contribution ahead of the academic-English study, and is the case in which the three evidence tiers are being developed.

The project's human/AI governance model — human owns pedagogy, AI executes under review — is formalised in a High-Level Design (HDD), a highly-evaluated Learning Experience Architecture artefact that evidences enterprise-level instructional-design skill (accessibility, measurement, modularity, risk/handoff) and anchors the environment's GOVERNANCE layer.

The PhD applicant brings 20+ years in university and upper-secondary (bachillerato) language education and instructional design — a Summa Cum Laude Bachelor's degree in Teaching and Education (English), curriculum specialisation, ranked first in class (18.975/20 — equivalent to 9.07/10 on the Spanish scale) — Universidad de Los Andes, Mérida, Venezuela (2004) — and an MA in Languages and Technology (UPV, GPA 9.4) — a final-degree project awarded an honorific distinction and published in an indexed journal (Eslava-Graterol, 2005, Kaleidoscopio, 2(3), 51–56; ISSN 1690-6054; indexed in CLASE, UNAM), professional instructional-design credentials (UC Irvine, 2025; University of Cambridge, 2025), and an acknowledgement in a peer-reviewed Elsevier article for "valuable comments on the manuscript's English and scientific content" (Sotelo-Casas et al., 2019; acknowledged as A. Z. Eslava de Morales) — situating the Design-Based Research in authentic teaching and scientific-writing contexts.

3.4 Point of departure — the proposed MLLDE framework. MLLDE nests established frameworks; the synthesis, targeted at mobile-native delivery, is the contribution, and each framework remains attributed to its author.

MLLDE – Mobile Language Learning Design Environment (mobile-native)

└ DESIGN

SAM → G-CLIL/IP0 (Casañ-Pitarch) → 4Cs → ESA → CTL/REACT → mobile UX + dual-coding

└ AI-LITERACY

Hyland (2005) metadiscourse baseline → human-vs-LLM signature → rhetorical agency [①]

└ TUTORING

Architecture Voice Design: language-based GPT mentor 'PhD Chema' + NEW acoustic

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pronunciation channel as a first-class assessed channel · pragmatics · lexical approach

└ MEASUREMENT

Measurement Module (Measure≠Decide; 3 evidence tiers) – WRITING (MET00L/Hyland) + SPEAKING (ASR·GOP)

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→ CEFR → xAPI (IEEE 9274.1.1-2023) → LRS; deterministic engine; dual telemetry logs the AI's teaching act

└ GOVERNANCE

EU AI Act (Art. 50 + record-keeping) · EAA/WCAG · GDPR-by-design · UDL – human owns pedagogy

4. References (cited in §3)

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Added — voice measurement, feedback & training-evaluation strand: - International Phonetic Association. (1999). *Handbook of the International Phonetic Association*. Cambridge University Press. - Kirkpatrick, J. D., & Kirkpatrick, W. K. (2016). *Kirkpatrick's four levels of training evaluation*. ATD Press. - Krashen, S. D. (1985). *The input hypothesis: Issues and implications*. Longman. - Phillips, J. J., & Phillips, P. P. (2016). *Handbook of training evaluation and measurement methods* (4th ed.). Routledge. - Schmidt, R. W. (1990). The role of consciousness in second language learning. *Applied Linguistics*, 11(2), 129–158. <https://doi.org/10.1093/applin/11.2.129>

Note. Entries verified against the MA thesis reference list (metadiscourse/AI-literacy apparatus) and public catalogues. A small number of items for books/standards (Coyle et al., 2010; Kasper & Rose, 2001; IEEE 9274.1.1-2023) are not issued as DOIs and are cited by publisher/URL per APA 7.

5. Hypotheses and objectives

Rationale & starting hypothesis. Because LLM text reproduces the scaffolding of language while flattening the interpersonal dimension (writing), and because AI pronunciation evidence is not interoperable (speaking), a single, ethically governed, mobile-native, teacher-authorable environment can (a) build learners' AI literacy and rhetorical agency and (b) make speaking measurable in CEFR-aligned, interoperable terms. **Starting hypothesis:** an MLLDE-designed mobile-native intervention improves AI-critical writing competence and yields valid, interoperable pronunciation evidence more effectively than non-integrated baselines.

General objective. To develop, formalise, and empirically validate MLLDE as an ethical, AI-literate design-and-measurement environment for mobile-native language applications.

Specific objectives (realistic for the thesis duration). 1. Formalise MLLDE — constructs, layers (design · tutoring · measurement · governance), and the mobile-native authoring schema for a **curriculum that integrates writing and the speaking / voice modality**, each with its own anchored measurement. 2. Instantiate/refine

Academic Writing Journey (AWJ) — contribution ① — extending Objective 4 of the MA thesis (corpus profiles → AI-literacy materials), taken mobile-native. 3. Refine the mobile-native gamified prototype as MLLDE's first demo (engine + data authoring). 4. Run the AWJ pilot with pre-/post-metadiscourse measures (the MA thesis's fourth future-research line). 5. Contribution ② — speaking: build Architecture Voice Design — extend the AWJ language-based custom-GPT agent *PhD Chema* with a first-class acoustic pronunciation channel (ASR/GOP) that its first version lacks — and the Measurement Module — a CEFR-aligned xAPI Profile (persistent identifiers, controlled vocabulary, acoustic ↔ CEFR mappings, pedagogical interpretation, interoperable records; dual telemetry logging the AI's teaching act). Develop pronunciation measurement across **three evidence tiers** — Tier 1 (rubric over ASR, in production), Tier 2 (instrumented vendor scoring), Tier 3 (custom GOP with a per-level IPA inventory) — under a **Measure ≠ Decide** engine and an evidence-honesty policy (intelligibility, not native accent). 6. Isolate mechanisms (human-vs-LLM exercises; answer-withholding AI scaffolding; animated hints). 7. Test replicability across levels/destinations/languages (design effort + fidelity). 8. Produce design guidelines, an educator authoring method, and a responsible-AI governance model.

Research questions & hypotheses. - RQ1 (design). What principles constitute an effective mobile-native MLLDE across writing and speaking? - **RQ2 (writing).** How does AWJ affect metadiscourse control and AI-critical writing vs. a baseline? — **H1:** larger pre-/post gains in interpersonal metadiscourse and human-vs-LLM discrimination. - **RQ3 (mechanisms).** Effects of human-vs-LLM exercises, answer-withholding scaffolding, animated hints? — **H2:** answer-withholding scaffolding raises learner production; **H3:** animated same-image hints improve recall / lower cognitive load. - **RQ4 (speaking, by tiers).** Can voice/acoustic evidence be formalised as a valid, interoperable CEFR-aligned xAPI Profile **across three evidence tiers** (rubric-over-ASR → instrumented → custom GOP), and what does validity gain at each tier? What is the defensible **intelligibility floor** per tier? — **H4:** xAPI-mapped acoustic scores align with expert CEFR ratings and improve over a no-feedback baseline; each tier adds signal without breaking the previous. - **RQ4b (measurement validity & honesty-as-method).** Does an LLM-applied CEFR rubric over the ASR transcript (Tier 1) yield bands that **agree with expert human raters** and remain **stable across model sizes**; and does the **evidence-honesty policy** — a **deterministic engine** (not an LLM) assigns the band, each claim declares its tier, unmeasured skills show "—", and all surfaces unify on a single LRS truth — prevent over-claiming and hallucinated scores? — **H6:** Tier-1 LLM–human agreement is acceptable and stable, and surface-unification resists hallucination. - **RQ5 (transferability).** Is MLLDE replicable across implementations, CEFR levels, and languages? — **H5:** a shared engine authors new instances with lower effort and preserved fidelity.

6. Methodology

Approach. Design-Based Research (Anderson & Shattuck, 2012) with build cycles following the Successive Approximation Model (Allen & Sites, 2012: Preparation → Alpha → Beta → Gold), already used for AWJ and the mobile-native prototype. Tasks and expected results:

Task	Activity	Expected result
T1	Formalise MLLDE; expert (Delphi) validation of framework, mobile-native schema & xAPI Profile	Validated framework + schema
T2	Refine AWJ & the mobile-native prototype — including AWJ Levels 5 & 6 (Output: writing & speaking/conference pitch) rebuilt in a mobile-native environment as the host for the acoustic module	Gold mobile-native prototypes
T3	AWJ pilot: pre-/post-metadiscourse measures + human-vs-LLM task	Writing/AI-literacy outcome data
T4	Mechanism experiments (scaffolding; hints)	Causal mechanism evidence
T5	Extend the language-based mentor <i>PhD Chema</i> with an acoustic pronunciation channel (ASR/GOP) + build the Measurement Module on the mobile-native AWJ Levels 5–6; validate acoustic xAPI vs. expert CEFR ratings	Interoperable pronunciation evidence

Task	Activity	Expected result
T6	Replication: new levels/destinations/languages; LRS interoperability	Replicability + fidelity data
T7	Synthesis; guidelines; authoring method; governance model	Design principles + thesis

Participants & recruitment. The primary sample is **adult learners of academic English (B2 → C1)** — the population AWJ is designed for — recruited from a large, readily available cohort (~**300 learners per year**) through the co-director, Prof. Casañ-Pitarch, giving an ecologically valid, well-powered pool for the writing / AI-literacy pilot (T3) and for the **mobile-native academic-English application** — AWJ taken mobile-native, with a **voice mode and a built-in assessment system** — hosting the speaking / pronunciation module (**Architecture Voice Design** + the **Measurement Module**: CEFR-aligned, xAPI pronunciation assessment; T5). During the PhD, this mobile-native academic-English app — voice mode **plus** its assessment system — is the primary built-and-tested artefact. The **replication study (T6)** then extends the method to **Spanish** (A1–A2, via the *Language Passport* mobile prototype) and to further CEFR levels and languages, testing cross-language transferability. Sample sizes, sampling, and any Spanish-cohort recruitment via UPV language services will be finalised with the directors and the ethics board.

The Measurement Module and the three evidence tiers. Speaking evidence is developed under a **Measure ≠ Decide** architecture: three observing LLM agents describe the interaction while a **deterministic engine** (code, not a fourth LLM) computes the 0–10 CEFR band ($\text{Global} \times 5 + \text{analytics} \times 2 \div \text{pool} \times 10$) and emits a single xAPI statement; a **measurement stream** (once per session) is firewalled from a **pedagogy stream** (per turn, which never scores). Pronunciation is built in three tiers — **Tier 1** a CEFR rubric over the ASR transcript (in production), **Tier 2** instrumented vendor pronunciation scoring (e.g. Azure Pronunciation Assessment), **Tier 3** a custom Goodness-of-Pronunciation model with a per-level **IPA inventory** (Witt & Young, 2000) — each declaring its `measure_model` for comparability and governed by an **evidence-honesty policy**: a claim never exceeds its tier's precision, unmeasured skills show "—", accepted variants (e.g. *seseoldistinción*) never lower a score (IPA, 1999), and an **acoustic gate** (ASR-confidence 0.70) turns low-confidence turns into non-scoring incidents. Feedback is dosed by a **noticing scale (0–4)** matched to CEFR level (model A2–B1, elicit B2+; Schmidt, 1990) within an *i+1* logic (Krashen, 1985).

Tier-1 validity study. LLM-rubric bands are checked against **expert human ratings** (agreement), for **test–retest stability**, and for **sensitivity to model size**; an **integrity audit** (panel + agent review, with an "H0 watchdog" that flags any descriptor rendered as a band) verifies that no surface over-claims. Stakeholder reports (learner · teacher · HR) use a **Kirkpatrick / Phillips** evaluation frame, with optional **ESCO** alignment.

Instruments & data. *Writing*: CEFR writing rubric; METOOL-based metadiscourse pre-/post measures (Carrió-Pastor, 2020); human-vs-LLM discrimination task. *Speaking*: acoustic measures (phoneme accuracy, stress, rhythm, intonation, speech rate, fluency, intelligibility) via ASR/GOP, mapped through the xAPI Profile and validated against expert CEFR phonological-competence ratings. *Shared*: self-efficacy/motivation scales; app telemetry + xAPI statements (dual telemetry: learner performance + AI teaching act); System Usability Scale; cognitive-load measures; interviews, think-aloud, expert reviews. *Analysis*: mixed methods — inferential statistics (pre/post, between-condition) triangulated with thematic analysis.

Critical evaluation of difficulties & contingency plan. (i) Single-coder tagging (MA limitation) → inter-rater reliability and test–retest checks. (ii) Recruitment/sample size → partner programmes; within-subject designs; pre-registered success criteria. (iii) AI-mentor reliability → constrained, answer-withholding prompts; human review; AI as supplementary, not a single point of failure. (iv) xAPI/LRS interoperability risk → adopt the established IEEE 9274.1.1-2023 specification; expert validation before scale-up. (v) Ethics/regulation → GDPR-by-design (pseudonymised, client-side where possible), EU AI Act (Art. 50 transparency + record-keeping/human-oversight), EAA/WCAG accessibility; UPV ethics approval before data collection.

7. Resources

UPV library and databases; the METOOL tagger (Carrió-Pastor, 2020) and the applicant's MA corpus/workflow; existing mobile-native prototypes (AWJ and a gamified mobile prototype) and design toolkit (Canva, Articulate 360

Rise, HTML/PWA); LLM conversational tutors (e.g., Anthropic's Claude, and other GPT-family models) — the core AI-literacy resource, powering the AWJ language-based conversational mentor *PhD Chema* (a custom GPT), the answer-withholding scaffolding, and the human-vs-LLM comparison exercises, used under the responsible-AI governance model; ASR/GOP speech-assessment tooling and audio capture on mobile devices; a conformant Learning Record Store (Veracity) for xAPI (IEEE 9274.1.1-2023) records and dashboards, plus a vendor pronunciation-assessment service (e.g. Azure) for the instrumented tier; statistical software (e.g., R/JASP) and qualitative-analysis software; access to learner cohorts via UPV language services / partner programmes. No specialised equipment beyond mobile devices is required.

8. Timeline and impact

Timeline (linked to §6 tasks).

Year	Tasks	Milestones
1	T1–T2	Framework paper; validated mobile-native schema; xAPI Profile v1; ethics approval
2	T3–T5	Outcome + mechanism papers; AWJ pilot results; acoustic xAPI validation
3	T6–T7	Replicability + interoperability papers; guidelines; thesis; defence

(Part-time schedule extends this to ~5 years.)

Measurement roadmap (evidence tiers). **Tier 1** — in production and verified (live LRS; interim measurement on a small model, declared per session). **Tier 2** — planned: integrate vendor pronunciation assessment to light up the Pronunciation band (today shown as “—”). **Tier 3** — research: a per-level IPA inventory + a custom Goodness-of-Pronunciation model (the phonemic contribution of the PhD). *Cross-cutting:* validate noticing uptake, and the reliability of LLM-rubric scoring by model size.

Utility & impact. This research sits at the crossroads of four fields — **AI in education** (AI literacy as a designed outcome), **language assessment** (CEFR-anchored, xAPI-interoperable measurement of writing and speaking), **applied NLP / human–AI interaction** (a responsibly scaffolded conversational tutor), and **responsible AI** (Article 50 transparency, human oversight, and GDPR-by-design); its contribution is their integration into a single, measurable, teacher-authorable environment. *Scientific:* a mobile-native environment framework, empirically grounded design principles, and a CEFR-aligned xAPI Profile carrying both metadiscourse (writing) and acoustic (speaking) evidence — contributing to AI-literacy, metadiscourse/EAP, MALL, CLIL, and learning-analytics research (national and international dissemination). *Social/educational:* AI-critical writing and speaking skills supporting equitable access to quality tertiary education and employability (SDG 4.3–4.4), in an accessible, teacher-authorable, phone-first form. *Economic/technical:* an engine + data authoring model that lets educators build mobile-native lessons without programming, and interoperable learner records reusable across LMS/LRS.

9. Training plan

- **Transversal / methodological courses:** research methodology, statistics for linguistics, corpus tools, learning analytics/xAPI, research ethics and open science (UPV doctoral school).
- **Domain seminars:** applied linguistics, CALL/MALL, gamification, generative AI in education (under the directors' guidance).
- **Conferences (dissemination):** e.g. EUROCALL, EATAW, AESLA and applied-linguistics / EdTech venues — one target per year.
- **Mobility:** a research stay at an international group in CALL / learning analytics / pronunciation assessment.
- **Publications:** framework paper (Y1); outcomes/mechanisms and acoustic-xAPI papers (Y2); replicability/interoperability paper (Y3).

This plan is developed under the direction of Prof. M^a Luisa Carrió-Pastor — for her invaluable contribution to the scientific field and applied linguistics — and the co-direction of Prof. Ricardo Casañ-Pitarch, to continue, through his LLM-related publications, the responsible integration of LLMs in language teaching. To be submitted to the Departamento de Lingüística Aplicada, Universitat Politècnica de València.

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